

No 20 011 901

Sonuç 1

$$R_1 = 1 \Omega$$

$$R_2 = 0.5 \Omega$$

$$R_3 = 9 \Omega$$

$$R_h = 0.5 \Omega$$

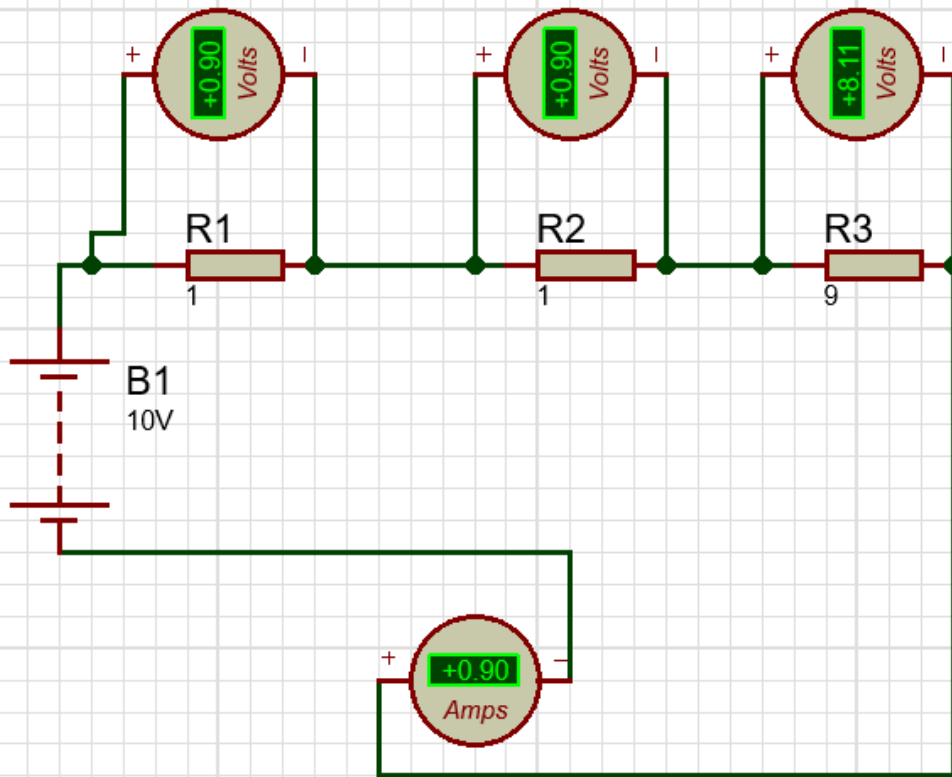
Durumlar	Seri	V	V ₁	V ₂	V ₃	V ₄	İ _ö	İ _h	R _{es} h	P _w (V _h · İ _h)
R _h çıkıyor	R ₁ , R ₂ , R ₃	10	0,9 V	0,9 V	8,1 V	—	0,9 A	0,9 A	11 Ω	9,1 W
R ₂ R _h çıkıyor	R ₁ , R ₃	10	0,99 V	—	8,99 V	—	0,99 A	1 A	10 Ω	10 W
—	R ₁ , R ₂ , R ₃ , R _h	10	0,83 V	0,83 V	7,44 V	0,83 V	0,83 A	0,83 A	12 Ω	8,33 W

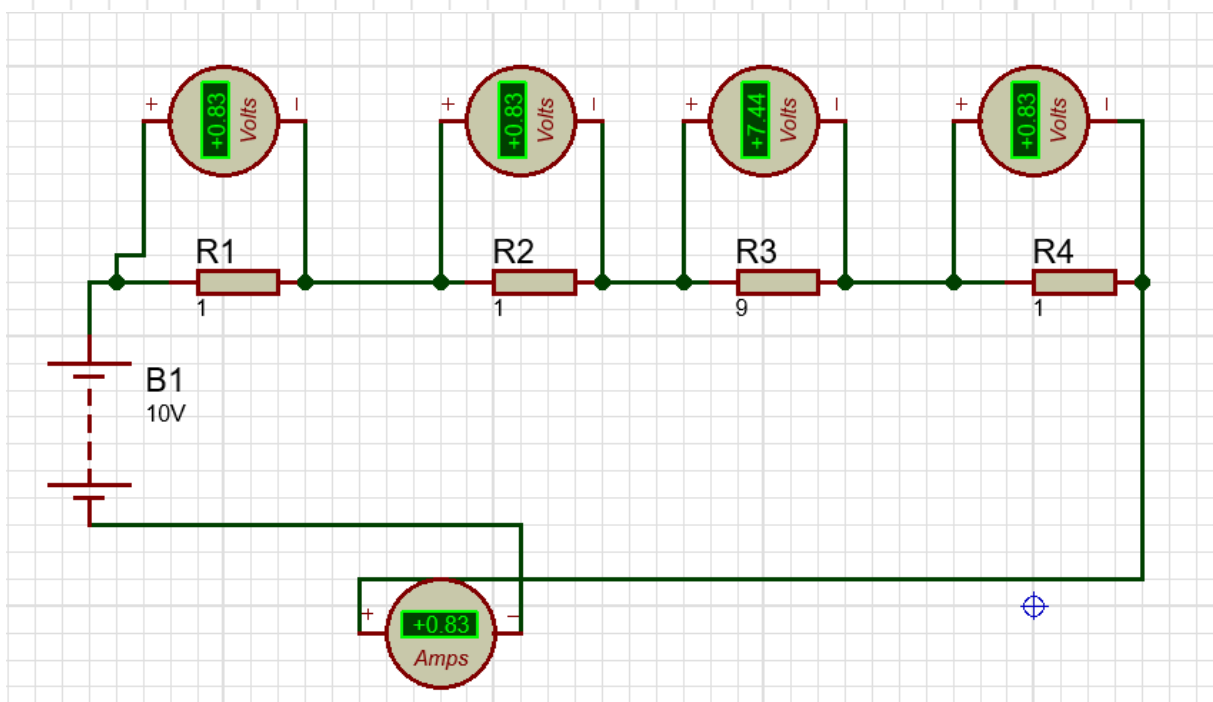
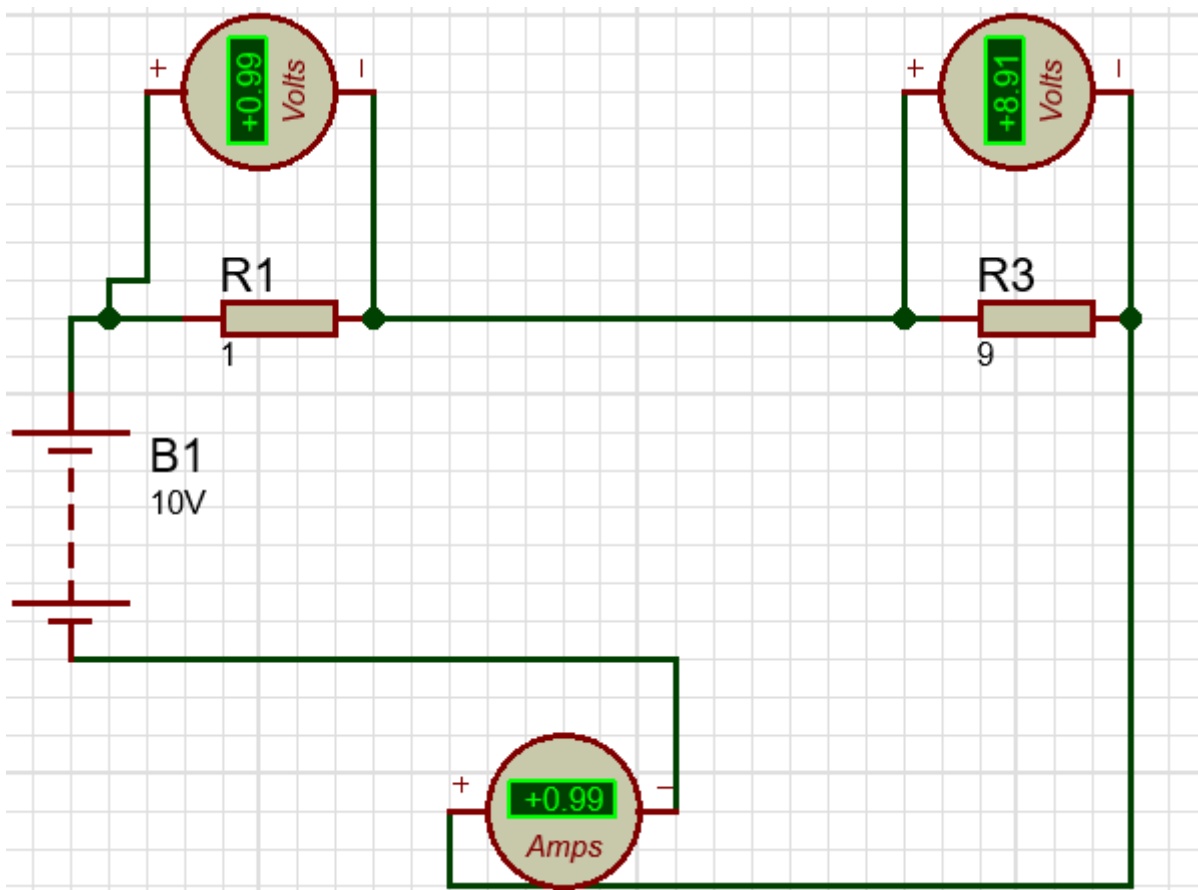
$$V = \Sigma R_{es} \cdot I$$

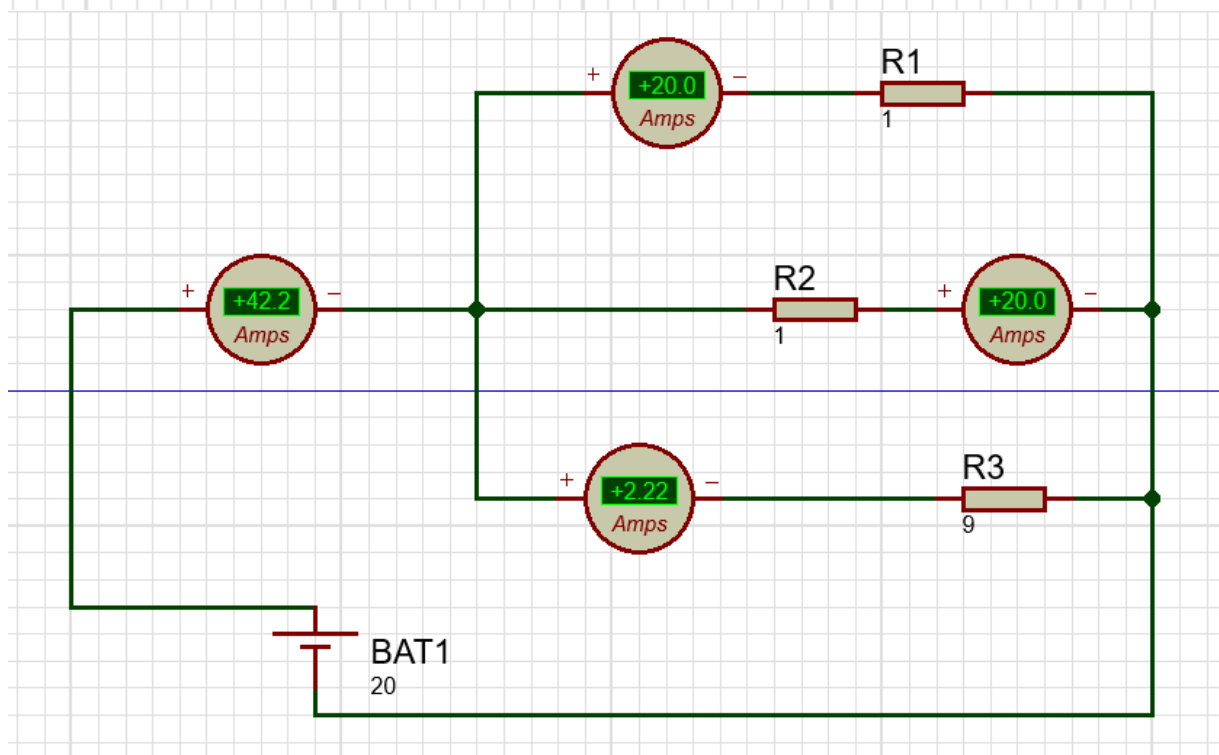
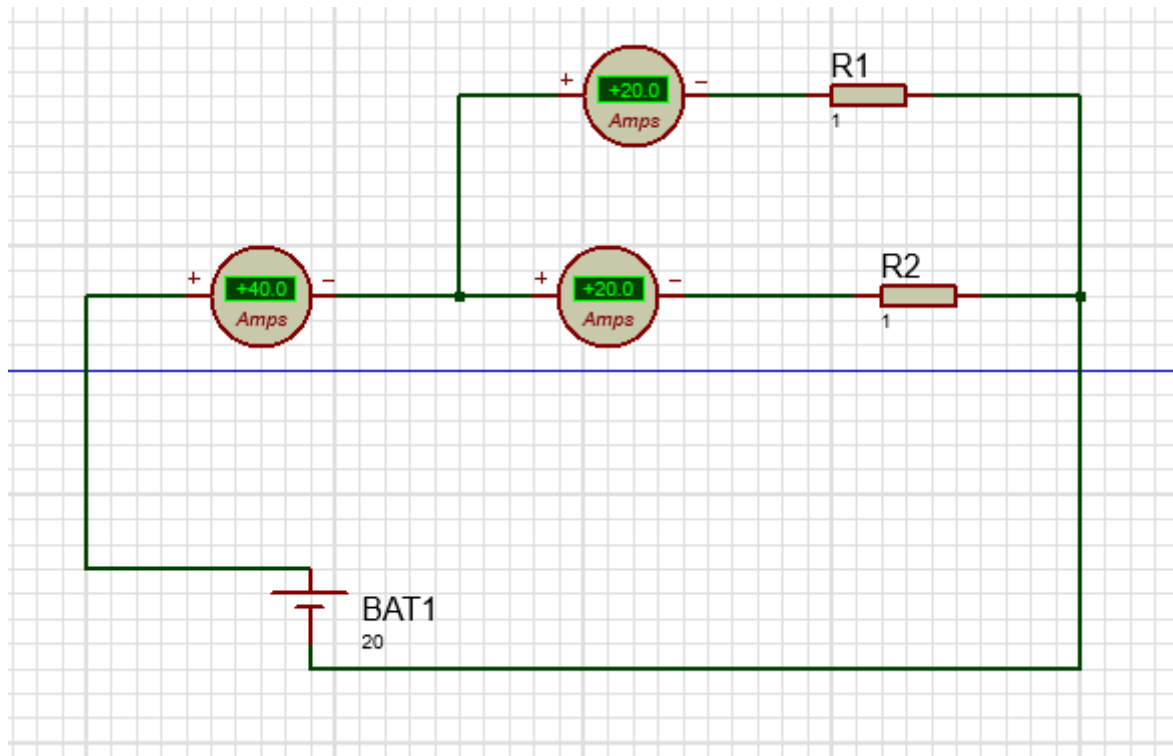
$$10 = \Sigma \cdot (9 + 1 + 1 + 1) \quad P = V \cdot \Sigma = 10 \cdot 0,833 = 8,33 \text{ W}$$

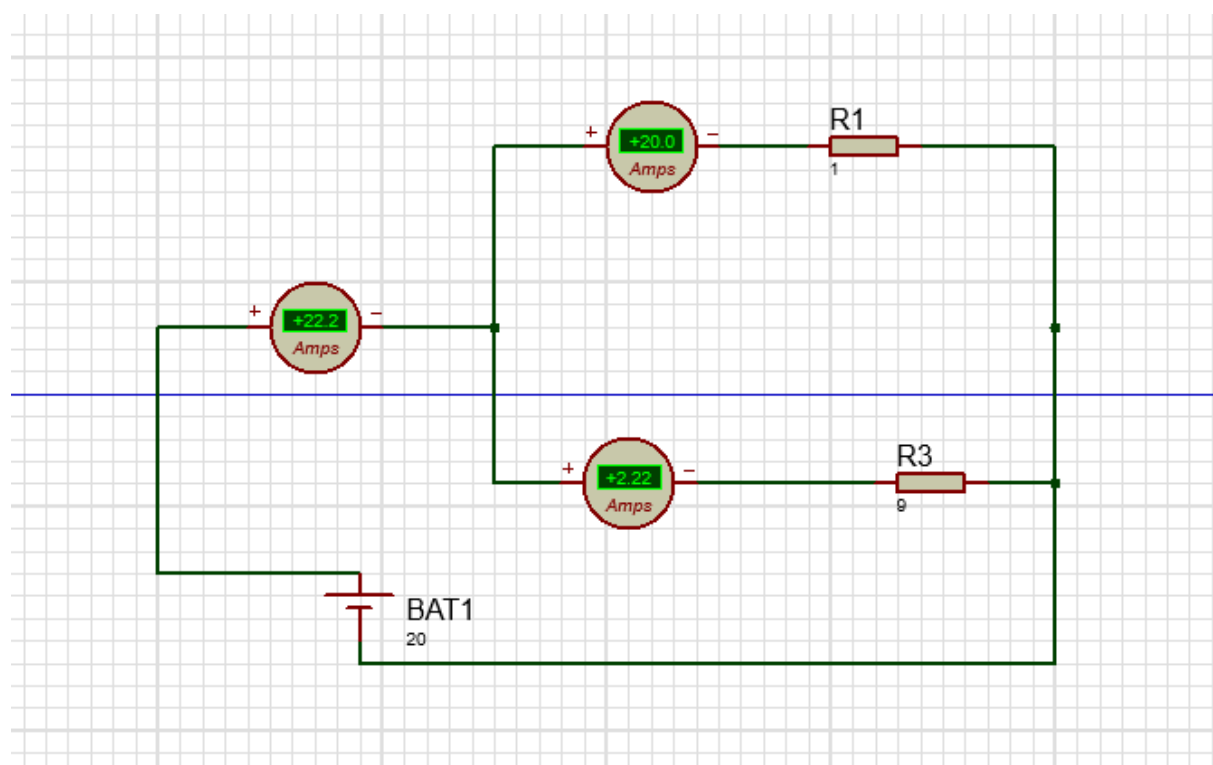
$$R_{es} = 9 + 1 + 1 + 1$$

$$\frac{10}{12} = 0,833 \text{ A}$$









$$V = I \cdot R_{eq} \quad 10 = I \cdot (1+9) \quad P = I \cdot V = 1 \cdot 10 = 10 \text{ W}$$

$$R_{eq} = 1+9$$

$$I = 1 \text{ A}$$

$$V = I \cdot R_{eq}$$

$$10 = I \cdot 11$$

$$P = I \cdot V = 0,91 \cdot 10 = 9,1 \text{ W}$$

$$R_{eq} = 9+1 = 10 \Omega$$

$$I = 0,91$$

Sen U 2

$$R_1 = 1 \Omega$$

$$R_2 = 1 \Omega$$

$$R_3 = 9 \Omega$$

Durum	Paralel	V	I_1	I_2	I_3	I_0	I_1	R_{eq}	P
R_3 9,1A	R_1, R_2	20	14,7 A	14,7 A	—	29,4 A	40 A	0,9 Ω	800 W
R_2 9,1A	R_1, R_3	20	18 A	—	2 A	20 A	22,2 A	0,9 Ω	444,4 W
—	R_1, R_2, R_3	20	14,5 A	14,5 A	11,2 A	34,7 A	42,2 A	0,43 Ω	844,4 W

$$V = I \cdot R_{eq}$$

$$20 = I \cdot \frac{9}{19}$$

$$R_{eq} = 0,473 \Omega$$

$$R_{eq} = \frac{1}{\frac{1}{1} + \frac{1}{1} + \frac{1}{9}} = \frac{9}{19}$$

$$I = 42,2 \text{ A}$$

$$P = 42,2 \cdot 20 = 844,4 \text{ W}$$

$$V = I \cdot R_{eq}$$

$$20 = I \cdot \frac{9}{10}$$

$$R_{eq} = 0,9 \Omega$$

$$R_{eq} = \frac{1}{\frac{1}{1} + \frac{1}{9}} = \frac{9}{10}$$

$$I = 22,2 \text{ A}$$

$$P = 22,2 \cdot 20 = 444,4 \text{ W}$$

$$V = I \cdot R_{eq}$$

$$20 = I \cdot R$$

$$P = I \cdot V = 40 \cdot 20 = 800 \text{ W}$$

$$I = 40 \text{ A}$$

$$R_{eq} = \frac{1}{\frac{1}{1} + \frac{1}{1}} = \frac{1}{2}$$